Adversarial Co-Evolution of RL and LLM Agents in Gin Rummy: A Phase 1–3 System Report with a Gold-Standard Yardstick

Nima Kelidari Mahdi Salmani Mohammadsaeed Haghi
{kelidari, salmanis, haghim}@usc.edu
University of Southern California

Abstract

We build a hybrid system in which a lightweight action-masked PPO agent and a strong-but-slow LLM opponent train against each other in Gin Rummy, without paying full LLM inference latency on every reinforcement-learning step. This report covers two phases. In **Phase 1** an action-masked PPO agent saturates against a random opponent: an eight-configuration hyperparameter sweep lands every run in a 98.3%–99.6% win-rate band, statistically indistinguishable at 1000-episode evaluation. Because further tuning against weak opponents is unproductive, **Phase 2** builds the infrastructure for LLM-in-the-loop training: a master/worker/cache inference stack with shared-filesystem service discovery and a suit-symmetry prompt cache, plus self-play and AlphaZero-style pool curricula for an in-distribution opponent. We report three operational findings: (i) OLMoE-1B is too weak to teach (it plays at chance), whereas Qwen2.5-7B-Instruct beats a random hero 5/5 and is a genuinely useful teacher; (ii) loading a 7B worker from the home NFS is 62× slower than from scratch/BeeGFS and must be avoided; and (iii) pool self-play diverged after ∼10M steps, dropping its win rate vs random from 98.5% to 85.8%. Finally we run a full **1.5M-step RL-vs-LLM fine-tune** (PPO warm-started from the self-play champion, up to 32 Qwen2.5-7B workers): episode reward stays flat at the knock value and the resulting agent matches—but does not beat—its progenitor. The honest conclusion is that a 7B opponent is *weaker* than our self-play champion, so it cannot lift a near-saturated policy. **Phase 3** confirms that reward shaping alone does not raise the gin rate, and a **gold-standard** expert agent (exact meld optimisation) beats our best RL agent 70% of the time, so the learned agents are far from optimal. This agent is a *benchmark yardstick only*: the project's aim is a framework that develops strong RL with an **LLM** as the guide (never the optimal solver) for tasks that lack any clear algorithm or gold standard, and **Phase 4** measures how close LLM-guided RL gets to that yardstick.

1 Goal and roadmap

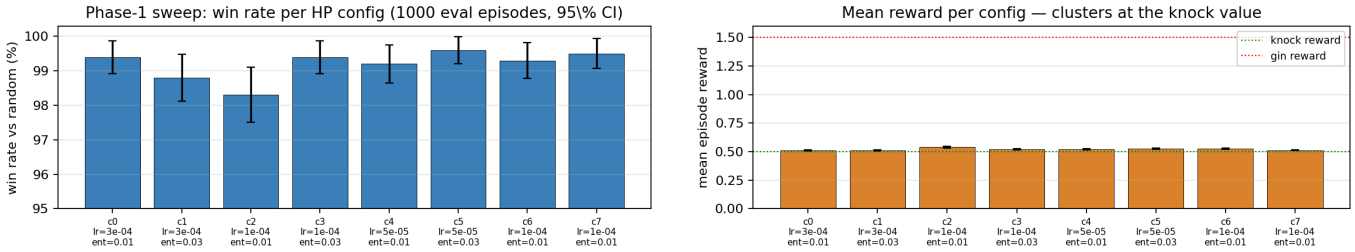
Gin Rummy is an imperfect-information card game requiring both short-horizon arithmetic (deadwood counting) and long-horizon planning (meld formation). Our roadmap has three phases: (1) train a strong RL backbone against weak opponents; (2) train that backbone against an LLM opponent that is strategically richer but operationally far slower; (3) let the two co-evolve. We use action-masked PPO [4, 1] on the PettingZoo [5] `gin_rummy_v4` environment (wrapping RLCard [6]) with Stable-Baselines3 [3]; masking sets illegal-action logits to $-\infty$ inside a custom actor–critic policy. We first fixed an upstream bug in which PettingZoo silently reverts the configured `knock`/`gin` rewards on every seeded reset (PettingZoo PR #1312); all results below use the corrected rewards (knock = 0.5, gin = 1.5).

2 Phase 1: the RL backbone saturates against random

We sweep eight configurations: a 3×2 grid over learning rate $\{3{\cdot}10^{-4}, 10^{-4}, 5{\cdot}10^{-5}\}$ and entropy coefficient $\{0.01, 0.03\}$, plus two ablations (rollout length 1024; 10 epochs). Each trains for 2×10^6 steps with 96 parallel envs on a 64-core node and is evaluated over 1000 deterministic games vs `RandomAgent`. Every configuration converges to the 98.3%–99.6% band (Fig. 1, Table 1); the spread is about one binomial 95% CI (±0.6 pp at p=0.99), so the configurations cannot be separated. The shared failure mode is informative: mean episode reward clusters at the knock value (0.5) rather than the gin value (1.5) (Fig. 1b), i.e. the agent takes the certain knock instead of holding for a gin—a risk/reward decision that only a *thinking* opponent can teach. This is what motivates Phase 2.

3 Phase 2: infrastructure for LLM-in-the-loop training

A single PPO rollout fires up to ∼50k opponent queries; at 0.5–3 s per 7B call, a naïve loop takes hours. We decouple inference from RL with a three-tier stack. **Workers** (one GPU each, HF backend) load the model and register themselves in a shared-filesystem registry. The **master** (CPU) discovers live workers, load-balances, and owns a prompt cache. The RL **client** (96 env subprocesses) reaches the master through the existing Ollama-compatible API, so no RL code changes. We chose the HuggingFace backend because the installed vLLM [2] 0.8.5 is incompatible with



(a) Win rate vs random.

(b) Mean reward.

Figure 1: Phase-1 sweep. All eight configs saturate vs random (a); reward sits at the knock value, far below gin (b), exposing the improvement headroom.

Table 1: Phase-1 leaderboard (top rows) and Phase-2 opponent panel (bottom). Win/loss over 1000 deterministic eval games unless noted.

Agent / config	opponent	win%	loss%	note
PPO cfg5 (lr 5e-5, ent .03)	random	99.6	0.4	best of sweep
PPO cfg7 (lr 1e-4, 10 epochs)	random	99.5	0.5	ablation
PPO cfg0 (lr 3e-4, ent .01)	random	99.4	0.6	baseline
PPO cfg3 (lr 1e-4, ent .03)	random	99.4	0.6	
PPO cfg6 (lr 1e-4, n_s=1024)	random	99.3	0.7	ablation
PPO cfg4 (lr 5e-5, ent .01)	random	99.2	0.8	
PPO cfg1 (lr 3e-4, ent .03)	random	98.8	1.2	
PPO cfg2 (lr 1e-4, ent .01)	random	98.3	1.7	worst of sweep
Self-play (3M, vs frozen run_5)	run_5	61.1	38.9	beats progenitor
Self-play (3M)	random	98.7	1.3	stays dominant
Pool champion (12M)	random	85.8	14.2	*diverged*
Pool champion (12M)	run_5	62.0	38.0	
OLMoE-1B opponent	random	≈50	—	plays at chance
Qwen2.5-7B opponent	random	**100**	0	5/5 (small N)
LLM-finetuned PPO (1.5M)	random	99.0	1.0	retains competence
LLM-finetuned PPO (1.5M)	self-play champ	43.3	56.7	*no net gain*
LLM-finetuned PPO (1.5M)	winrate / reward	55.3 / 61.3	—	still beats weak agents
Gold-standard	random	99.5	0.5	near-optimal
Gold-standard	self-play champ	**70.2**	29.8	beats best RL
Gold-standard	LLM-finetuned	71.7	28.3	

transformers 5.7. The cache uses a *suit-symmetry canonical key*: Gin Rummy is invariant under suit relabeling, so suit-equivalent states share one entry and the chosen action is de-canonicalized on return.

Two infrastructure findings. First, loading a 7B worker from the home NFS runs at ∼11 MB/s and takes ∼28 min—long enough to blow the health-check timeout. Staging weights on scratch/BeeGFS cuts this to 27 s, a 62× speedup (Fig. 3c); this is mandatory when launching dozens of workers. Second, the in-game opponent model is selected purely by the worker's loaded weights—the master ignores the client's model string—so swapping opponents is a one-line preset change.

4 Opponents: self-play, pool, and the LLM

Self-play. Fine-tuning the Phase-1 champion against a *frozen* copy of itself for 3M steps yields 58–61% vs that progenitor while holding ∼98.7% vs random (Fig. 3a)—one clean generation of improvement.

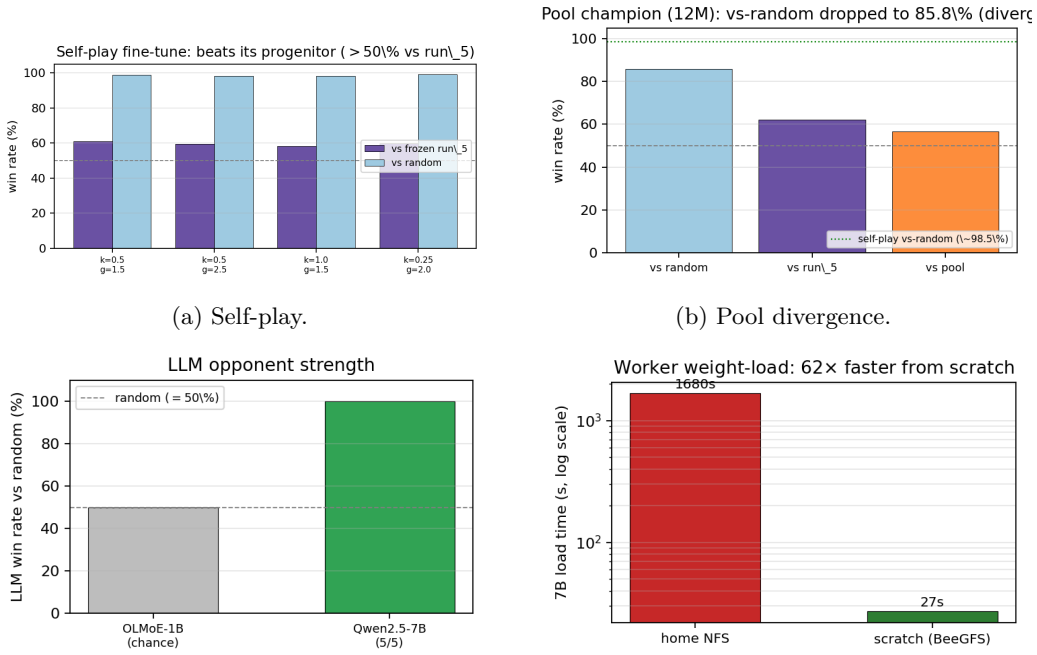
Pool (AlphaZero-style). Training against a growing pool of past checkpoints ran to 12M steps but *diverged* after ∼10M: vs-random win rate fell from ∼98.5% to 85.8% (Fig. 3b). We keep the pre-divergence checkpoints; the production opponent reverts to self-play.

LLM opponent. OLMoE-1B cannot follow the action-format prompt—most responses fail to parse and fall back to a random move, so it plays at chance and cannot teach. Qwen2.5-7B-Instruct, by contrast, beats a random hero 5/5 (Fig. 3d): a genuinely competent teacher. Its chain-of-thought prompt costs ∼16 s/call (1024 output tokens); for

training we switch to a terse prompt that emits only the action and cap output tokens, cutting latency by an order of magnitude.

Human play. A zero-dependency web client (HTML/JS, Fig. 2) lets a human play any of the four trained opponents (Phase-1, self-play, pool, reward-shaped) with 3-D card animations; it reads opponent state from RLCard and was validated in headless Chromium.

Figure 2: The human-vs-RL web client (debug view, opponent hand revealed). The player drafts and discards against any trained agent; the board animates draws, reveals, and discards. The same `LLMAgent` path that the RL loop uses also drives an optional LLM opponent here.



(a) Self-play.

(b) Pool divergence.

(c) LLM strength.

(d) Worker load.

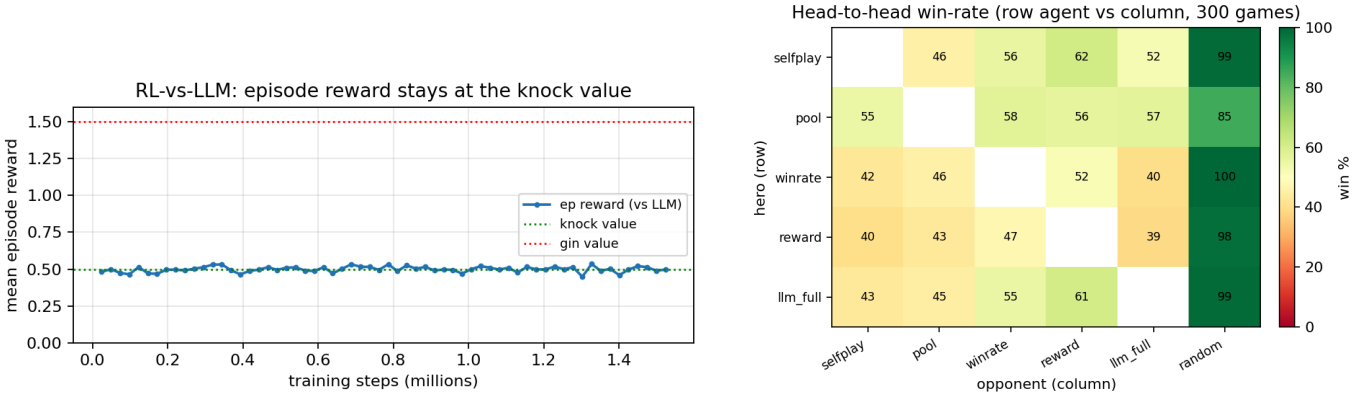
Figure 3: Phase-2 results. (a) self-play beats its progenitor; (b) the pool champion regressed vs random after divergence; (c) Qwen2.5-7B is a competent opponent where OLMoE-1B is not; (d) scratch staging is 62× faster.

5 Full RL-vs-LLM run: results, limitations, next steps

We then ran the real experiment: PPO warm-started from the self-play champion, 96 env subprocesses, fine-tuned for **1.5M steps** against the master + a pool of up to 32 Qwen2.5-7B workers (the scheduler granted ∼14 under the shared GPU quota; the pool is elastic). The serving stack is the headline infrastructure win: the terse prompt + a 64-token cap cut effective per-call latency from ∼16 s to ∼0.4 s, the master sustained ∼32 load-balanced queries/s (∼500× a

single-stream chain-of-thought loop), and PPO held ∼39 env-steps/s—so a rollout completes in minutes and the whole run finished in a few hours.

The learning result is negative and clean. Across all 1.5M steps the mean episode reward stays pinned at the knock value ≈ 0.49 and never trends toward gin (1.5) (Fig. 4a). The fine-tuned agent *retains* 99.0% vs random and still beats the weaker reward-shaped agents, but sits at 43.3% vs the self-play champion it started from (Fig. 4b, Table 1) —no net gain. The cause is structural: Qwen2.5-7B with a terse prompt is a *weaker* player than our self-play champion, so it provides no gradient toward stronger play; training a near-saturated policy against a weaker teacher holds it in place (and slightly perturbs it). The suit-symmetry cache also stayed near 0% hit-rate—opening states are too diverse to recur at this scale—so throughput is purely worker-bound.



(a) Reward flat over 1.5M steps.

(b) Head-to-head win-rate.

Figure 4: Full RL-vs-LLM run. (a) episode reward never leaves the knock value, so the LLM teacher did not raise the gin rate; (b) the fine-tuned agent (`llm_full`) lands at champion level—beating the weak agents and random, roughly even vs the champion it came from.

Limitations. (i) The teacher is weaker than the trainee—the central reason no improvement appears; (ii) the terse prompt trades opponent quality for throughput, and we have not measured a strong-prompt / larger-model teacher; (iii) single seed per run and a small (N=5) LLM-strength sample; (iv) the prompt cache yields no speedup at current volume; (v) pool self-play diverged past 10M; (vi) the midterm's negative DAgger and dense-reward-shaping results remain code-unaudited. Phase 3 (below) adds three-seed evidence on (iii); a stronger teacher for (i)/(ii) is deferred.

Phase 3 (first result: reward shaping). We tested whether reward shaping can close the gin headroom. Resuming the champion, we ran six reward configs (gin incentive escalating from baseline to gin-only) across three seeds, 8M self-play steps each, scoring the gin rate under *fixed* eval rewards so configs stay comparable. **The result is flat and negative.** The gin rate vs random holds in the 1.1 to 1.5% band for every config, statistically identical from the baseline (knock 0.5, gin 1.5: 1.40%) to the most extreme incentive (knock 0.25, gin 4.0: 1.40%), while win rate stays near 98% and the agents still beat their progenitor (57 to 60%). Terminal reward shaping is not the lever: the bottleneck is exploration and credit assignment, not the size of the incentive, since a near-saturated policy seldom visits the long held-card sequence a gin requires. The next step is dense, potential-based shaping that rewards each unit of deadwood reduction. The stronger Qwen3-30B teacher (the co-evolution test) is deferred: at two GPUs per worker it did not schedule under the shared quota.

A gold-standard yardstick. To calibrate the learned agents we built an expert agent that melds optimally (exact deadwood minimisation), takes the up-card only when it lowers deadwood, and knocks as soon as it is legal. Over 600 seat-swapped games it beats every learned agent: random 99.5%, reward-shaped 79.8%, pool 75.5%, winrate 74.3%, LLM-fine-tuned 71.7%, and the self-play champion 70.2% (gold-vs-gold 51%, a sanity check). The learned agents are thus strong but far from optimal, mainly in melding and discard quality rather than the knock call. A gin-seeking variant gins 96% and wins 96% vs random but wins only 15% vs the champion, who knocks before the gin lands: going for gin pays only against weak opponents, which is why a self-play agent rationally knocks and why Phase 3 shaping did not move it. We use this agent strictly as a *benchmark yardstick*: it is never used to train, because the project targets tasks where no such optimal solver exists.

Phase 4: an LLM-guided RL framework. The project's real goal is a reusable framework, an environment plus an RL base plus an **LLM guide**, that develops a strong RL policy for tasks where no clear algorithm or gold standard

Figure 5: Gold-standard benchmark, 600 seat-swapped games each. Left: it beats every learned agent, including the self-play champion (70%). Right: holding for a gin wins 96% vs random but only 15% vs the champion, so going for gin pays only against weak opponents.

exists. Gin Rummy is the testbed where a gold standard happens to exist, so we can measure how close LLM-guided RL gets to it; in the target domains the yardstick would be absent. The training signal therefore comes from the LLM, never from the optimal solver. The first mechanism is *LLM-DAgger*: a chain-of-thought LLM proposes the move at states the RL agent visits, the RL agent imitates these labels as a prior and then improves by RL beyond the (imperfect) LLM. The open question Phase 2 raised, whether a *sub-optimal* LLM can still bootstrap a strong RL policy, is exactly what this measures, scored as win rate against the gold-standard yardstick and the champion.

Reproducibility. Sweeps run as SLURM arrays under `sweep/`; the LLM stack is three `slurm/` jobs (master, a scratch-staged worker array, and the trainer); the gold-standard agent is `agents/gold_standard_agent.py`, benchmarked by `sweep/bench_gold.py`; every figure is regenerated by `make_figures.py` from the JSON result files.

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